

$$Z(T(n)) = Z(2T(n-1)) + Z(n2^n)$$

$$X(z) = 2(X(z)(z^{-1}) + T(-1)) + \frac{2z}{(z-2)^2}$$

$$(X(z)) \left[1 - \frac{2}{z}\right] = 2(0) + \frac{2z}{(z-2)^2}$$

$$(X(z)) = \frac{2z}{(z-2)^2} \left(\frac{z}{z-2}\right) \Rightarrow \boxed{X(z) = \frac{2z^2}{(z-2)^3}}$$

Applying the inverse z transform:

$$\begin{aligned} Z(n(2^n)U(n)) &= \frac{2z}{(z-2)^2} \\ &= \frac{(2z)(z-2)}{(z-2)^3} = \frac{2z^2 - 4z}{(z-2)^3} \quad \text{--- (1)} \end{aligned}$$

By differentiation property,

$$\begin{aligned} Z(n^2(2^n)U(n)) &= -z \frac{\partial}{\partial z} \left(\frac{2z}{(z-2)^2} \right) \\ &= \frac{2z^2 + 4z}{(z-2)^3} \quad \text{--- (2)} \end{aligned}$$

$$Z(n(2^n)U(n) + n^2(2^n)U(n)) = \frac{4z^2}{(z-2)^3}$$

$$Z\left(\frac{1}{2}(n^2+n)(2^n)U(n)\right) = \frac{2z^2}{(z-2)^3}$$

So, the required function is: $\boxed{\frac{1}{2}(n^2+n)(2^n)}$